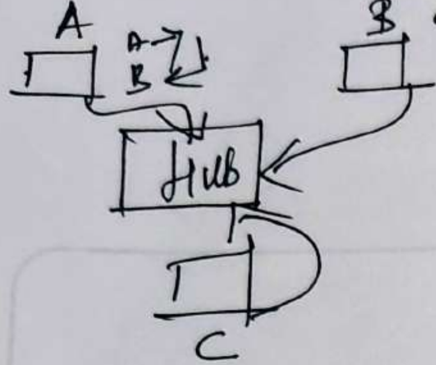
1, Explain behaviour of hub device



• Hub is a Hardware device used to connect computers to transmit and receive data. When data from A wants to Sent to B. It reaches the Hub and it immediately broadcast the data to B and C. As per ethics C drops the data & B receives it

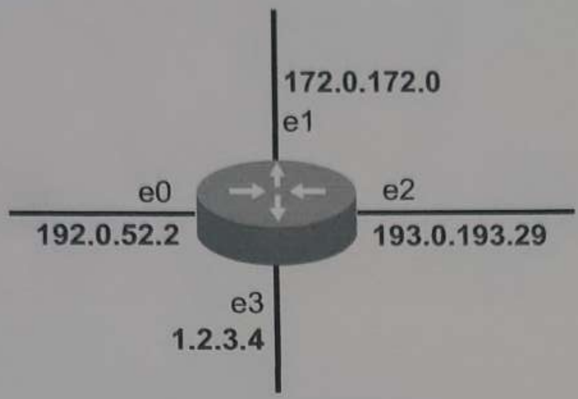
2, Explain behaviour of switch device

In Hub, two different data cannot travel Simultaneously and collission happens. The Connection is Half duplex.
Switch is a device used to connect two or more devices/ network together.
Switch maintains a mac table → it notes port no & destination mac. Switch can do both Broadcast and Multicast. It performs address learning first and the time duration of each data is 5 mins.

3, Explain behaviour of router device

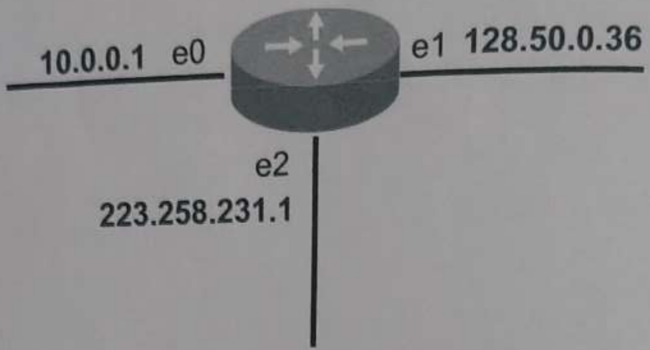
Router is device used to connect different networks.

It maintains a routing table → network id | prefix | port no
It reduces the problem of Single domain broadcast
It Unicast device

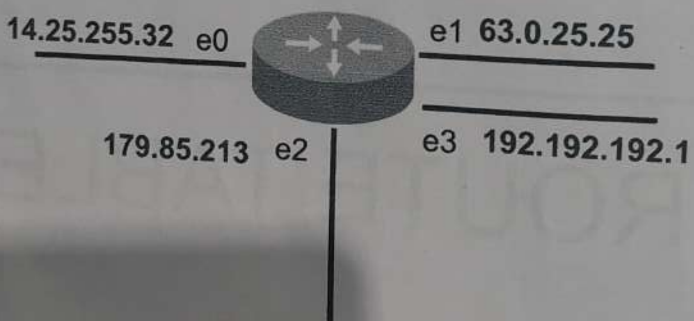


NETWORK	PREFIX	INTERFACE
192.0.52.0	/24	e0
172.0.0.0	/16	e1
193.0.193.0	/24	e2
1.0.0.0	/8	e3

Write directly connected routes



NETWORK	PREFIX	INTERFACE
10.0.0.0	/8	e0
128.50.0.0	/16	e1
223.258.231.0	/24	e2



NETWORK	PREFIX	INTERFACE
14.0.0.0	/8	e0
63.0.0.0	/8	e1
179.85.0.0	/16	e2
192.192.192.0	/24	e3